DELIGNE'S LETTER TO PIATETSKI-SHAPIRO

Princeton, March 25, 1973

Dear Piatetski-Shapiro,

Except at 2, I have now a good understanding at the bad primes of the ℓ-adic representations attached to modular forms (for $\mathrm{GL}(2, \mathbf{Q})$).

(A). Isogenies

Definition. The category of *elliptic curves up to isogeny* is obtained from that of elliptic curves by inverting isogenies.

I.e.,

(a) an elliptic curve E defines an elliptic curve up to isogeny $E \otimes \mathbf{Q}$
(b) $\mathrm{Hom}(E \otimes \mathbf{Q}, F \otimes \mathbf{Q}) = \mathrm{Hom}(E, F) \otimes \mathbf{Q}$

Hence, if F is a functor (ell. curves) $\to$ (...), and F(any isogeny) is an isomorphism, F makes sense for elliptic curves up to isogeny.

Notations:

$$\begin{aligned}
T_\ell(E) &= \varprojlim E_{\ell^n} \\
&\quad \text{(for } \ell \text{ prime to } p,\ E/k \text{ alg. clos. of char. } p, \\
&\quad \text{it is a free module of rank 2 on } \mathbf{Z}_\ell) \\
V_\ell(E) &= T_\ell(E) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell, \\
&\quad \text{(makes sense for ell. curves up to isogeny)} \\
\hat{T}_{p'}(E) &= \varprojlim_{(p,n)=1} E_n \\
\hat{V}_{p'}(E) &= \hat{T}_{p'}(E) \otimes_{\mathbf{Z}} \mathbf{Q}, \\
&\quad \text{a } \mathbf{A}^{f,p'} = \textstyle\prod'_{\ell \neq p, \infty} \mathbf{Q}_\ell\text{-module of rank 2} \\
&\quad \text{(makes sense for ell. curves up to isogeny)} \\
\mathbf{Z}_\ell(1) &= \varprojlim \mu_{\ell^n} \\
\mathbf{Q}_\ell(1) &= \mathbf{Z}_\ell(1) \otimes \mathbf{Q}_\ell \\
M(1) &= M \otimes \mathbf{Q}_\ell(1) \text{ or } M \otimes \mathbf{Z}_\ell(1)
\end{aligned}$$

The operator d is defined by

$$\begin{cases} d(E) = \mathbf{Z} \\ f \colon E \to F \implies d(f) \colon d(E) \to d(F) \text{ is } \deg(f) \text{ (0 if } f = 0) \end{cases}$$

Theorem.

(a) $d(E) \otimes \mathbf{Q}$ makes sense for elliptic curves up to isogeny. Notation: $d(E \otimes \mathbf{Q}) = d(E) \otimes \mathbf{Q}$.

(b) For E_0 an elliptic curve up to isogeny, the e_n-pairings, and their behavior under isogeny, enables one to define

$$\bigwedge^2 V_\ell(E_0) = d(E_0) \otimes \mathbf{Q}_\ell(1)$$

At p: Let us look at the case of supersingular curves of char. p; there, a substitute for $T_p(E)$ is the formal group of E (a height 2 dim. 1 formal group). The formalism of d has the following analogue:

(a) A height 2 dim. 1 formal group defines $d_p(F)$, a rank one module over $\mathbf{Z}_p$:

$$d_p(F) = \{u\colon F \to F^*\} \ (F^* = \text{Pontrjagin dual}), \text{ with } {}^t u = -u$$

(b) If $\tilde{F}/S$ is a deformation of F over S (local complete), and if $T_p(\tilde{F})$ is the corresponding local system on $S \otimes_{\mathbf{Z}_p} \mathbf{Q}_p$, then

$$\bigwedge^2 T_p(\tilde{F}) \cong d_p(F)(1)$$

(c) For E a supersingular elliptic curve, with corresponding formal group $\hat{E}$,

$$d(E) \otimes \mathbf{Z}_p \cong d_p(\hat{E}).$$

Recovering E: Let E_0 be a supersingular elliptic curve up to isogeny. An elliptic curve E with an isomorphism $\beta\colon E \otimes \mathbf{Q} \tilde{\to} E_0$ defines

(a) A "lattice" $\hat{T}_{p'}(E)$ in $\hat{V}_{p'}(E_0)$
(b) A "lattice" $d(E)$ in $d(E_0)$. Note that the p-aspect of it is determined by (a):

$$d(E) \otimes Z_\ell(1) \cong \bigwedge^2 T_\ell(E) \subset \bigwedge^2 V_\ell(E) = d(E_0) \otimes \mathbf{Q}_\ell(1)$$

Lemma. It amounts to the same to give either

(a) (E, β)
(b) The lattices $\hat{T}_{p'}(E) \subset \hat{V}_{p'}(E_0)$ and $d(E) \otimes \mathbf{Z}_p \subset d(E_0) \otimes \mathbf{Q}_p$

The reason why no more information is required at p is that, for supersingular E, the only degree p^k isogeny of source E is $E \to E^{(p^k)}$.

Variant. Let F_0 be a height 2 dim. 1 formal group law, up to isogeny. Then, to give F defining F_0 amounts to giving $d_p(F) \subset d_p(F_0)$.

(B). The fundamental local construction

Let:

$\overline{\mathbf{Q}}_p$	be an algebraic closure of $\mathbf{Q}_p$
$\overline{\mathbf{F}}_p$	be the alg. closure of $\mathbf{F}_p$, residue field of $\overline{\mathbf{Q}}_p$
k	be an alg. closure of $\mathbf{F}_p$, provided with a class modulo integral powers of Frobenius of allowed isomorphisms $k \cong \overline{\mathbf{F}}_p$. The class is denoted $\hom(k, \overline{\mathbf{F}}_p)$.
V_0	be a 2-dimensional vector space over $\mathbf{Q}_p$
E_0	be a supersingular elliptic curve up to isogeny over k
F_0	be the formal group law up to isogeny $/k$ defined by E_0.

To be able to use "transport de structure", I prefer not to take $k = \overline{\mathbf{F}}_p$ nor $V_0 = \overline{\mathbf{Q}}_p^2$.

1. **First construction.** Let $\sigma \in \hom(k, \overline{\mathbf{F}}_p)$ and $\beta \in \hom(d(E_0) \otimes \overline{\mathbf{Q}}_p(1), \bigwedge^2 V_0)$. Here, $\mathbf{Q}_p(1)$ is *relative to* $\overline{\mathbf{Q}}_p$:

$$\mathbf{Q}_p(1) = \mathbf{Q}_p \otimes_{\mathbf{Z}_p} \left(\varprojlim_n \text{group of } p^n\text{-roots of unity of } \overline{\mathbf{Q}}_p \right)$$

Then σ and β do define $E_0(\sigma, \beta) = (\sigma(E_0), \sigma(\beta))$ where $\sigma(E_0)$ is an elliptic curve up to isogeny over $\overline{\mathbf{F}}_p$, and $\sigma(\beta)$ is an isomorphism of one-dimensional vector spaces over $\overline{\mathbf{Q}}_p$

$$d(\sigma(E_0)) \otimes \overline{\mathbf{Q}}_p(1) \overset{\sigma}{=} d(E_0) \otimes \mathbf{Q}_p(1) \xrightarrow{\sim} \bigwedge^2 V_0.$$

Let ϕ be the Frobenius substitution $\phi\colon k \to k$, $x \mapsto x^p$. Then, for any elliptic curve E/k, $\phi(E)$ is $E^{(p)}$, and one disposes of the Frobenius isogeny $F\colon E \to E^{(p)}$. The diagram

$$\begin{tikzcd}
 & d(E) \arrow[rr, leftrightarrow, "\phi"] \arrow[d, equal] \arrow[ddl, equal] & & d\left(E^{(p)}\right) \arrow[d, equal] \arrow[ddl] \\
 & \mathbf{Z} \arrow[rr, equal, "p" near end] & & \mathbf{Z} \\
d(E) \arrow[rr, "F"] & & d\left(E^{(p)}\right) &
\end{tikzcd}$$

is commutative. In particular, F is an isomorphism $F\colon E_0 \tilde{\to} \phi(E_0)$, and the diagram

$$\begin{tikzcd}
d(E_0) \arrow[r, "\phi"] \arrow[d, equal] & d(\phi(E_0)) \arrow[d, "p"] \\
d(E) \arrow[r, "F"] & d(\phi(F_0))
\end{tikzcd}$$

is commutative. F hence induces an isomorphism

(1) $$F \text{ or } \sigma F\colon (\sigma(E_0), \sigma(\beta)) \xrightarrow{\sim} (\sigma\phi(E_0), \sigma(p^{-1}\beta))$$

Definition. D_p is the quotient of

$$\operatorname{Isom}(k, \overline{\mathbf{F}}_p) \times \operatorname{Isom}\left(d(E_0) \otimes \mathbf{Q}_p(1), \bigwedge^2 V_0 \right)$$

by the equivalence relation $(\sigma, \beta) \sim (\sigma\phi^k, p^{-k}\beta)$ for $k \in \mathbf{Z}$. It is a one-dimensional vector space over $\mathbf{Q}_p$.

Eq. (1) defines and isomorphism between $E_0(\sigma, \beta)$ and $E_0(\sigma', \beta')$ for $(\sigma, \beta) \sim (\sigma', \beta')$, and those isomorphisms form a transitive system of isomorphisms. They hence allow us to define $(E_0(\delta), \beta(\delta))$ for $\delta \in D_p$, where

(a) $E_0(\delta)$ is a (supersingular) elliptic curve up to isogeny over $\overline{\mathbf{F}}_p$, and
(b) $\beta(\delta)$ is an isomorphism $d(E_0(\delta)) \otimes \mathbf{Q}_p(1) \tilde{\to} \bigwedge^2 V_0$.

Remark. The following groups act on D_p (by "transport de structure")

(a) $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$, via its actions both on $\mathrm{Isom}(k, \overline{\mathbf{F}}_p)$ and $\mathbf{Q}_p(1)$. If the isomorphism of $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)^{\mathrm{ab}} = \mathbf{Q}_p^\times$ is normalized ($\pm$) so that Frobenius ϕ corresponds to *inverse of uniformizing parameter, then*

$$\sigma\delta = d(\sigma)^{-1} \cdot \delta.$$

(b) $\mathrm{GL}(V_0)$*, via its action on* $\bigwedge^2 V_0$*. One has*

$$g \cdot \delta = \det(g) \cdot \delta.$$

(c) $\mathrm{Aut}(E_0)$*, via its action on* $d(E_0)$*. We define*

$$H = \mathrm{Aut}(E_0),$$

it is the multiplicative group of a quaternion algebra ramified at p and at ∞*. One has*

$$g \cdot \delta = \mathrm{Nrd}(g)^{-1} \cdot \delta \textit{ reduced norm.}$$

Remark. Assume that a lattice $\Lambda \approx \mathbf{Z}_p$ has been chosen in $\bigwedge^2 V_0$. Then, $(E_0, \beta)/\overline{\mathbf{F}}_p$ as above define an elliptic curve p to p'-isogeny on $\overline{\mathbf{F}}_p$, corresponding to the lattice $\beta^{-1}(\Lambda)(-1) \subset d(E_0) \otimes \mathbf{Q}_p$ (p'-isogeny means isogeny of degree prime to p, *c.f., Lemma A.*

Variant. Let us start with F_0 instead of E_0. D_p is defined as before, using $d_p(F_0)$ instead of $d(E_0) \otimes \mathbf{Q}_p$, and is the same as before, via the isomorphism $d(E_0) \otimes \mathbf{Q}_p \cong d_p(F_0)$ ($F_0 = \hat{E}_0$). This time, D_p is acted upon by $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$, $\mathrm{GL}(V_0)$ and $\mathrm{Aut}(F_0) = H(\mathbf{Q}_p)$. The formulas are the same as before.

An element $\delta \in D_p$ defines

(a) $F_0(\delta)$, a (supersingular) formal group law up to isogeny on $\overline{\mathbf{F}}_p$
(b) $\beta(\delta)$, an isomorphism $d(F_0(\delta)) \otimes \mathbf{Q}_p(1) \tilde{\to} \bigwedge^2 V_0$

If Λ has been chosen in $\bigwedge^2 V_0$, one gets

(a) $F(\delta)$, a (supersingular) formal group law on $\overline{\mathbf{F}}_p$.
(b) $\beta(\delta)$, an isomorphism $d(F(\delta)) \otimes \mathbf{Z}_p(1) \tilde{\to} \Lambda$

2. **Second Construction.** I have now to define "vanishing cycle varieties" and vanishing cycle groups. Eventually, for K^0 and open compact subgroup of $\mathrm{SL}(V_0)$, and for $\delta \in D_p$, a scheme $V(K^0, \delta)$ over $\overline{\mathbf{Q}}_p$ will be defined. For K an open compact subgroup of $\mathrm{GL}(V_0)$ such that $K^0 = K \cap \mathrm{SL}(V_0)$, isomorphisms

$$V(K^0, \delta) \leftrightarrow V(K^0, \alpha\delta) \text{ (for } \alpha \in \det K^0 \subset \mathbf{Z}_p^\times)$$

are defined. For variable δ, the $V(K^0, \delta)$ will thus be a "local system" of schemes on D_p. The $V(K^0, \delta)$ are not of finite type, but of the type usual in the vanishing cycle theory. We won't need this, however. Here is a description of the $\overline{\mathbf{Q}}_p$-valued points of $V(K^0, \delta)$.

(a) δ provides us with $(E_0(\delta), \beta(\delta))$ on $\overline{\mathbf{F}}_p$.
(b) A point of $V(K^0, \delta)$ is an isomorphism class of systems consisting in
  (1) An elliptic curve up to isogeny on $\overline{\mathbf{Q}}_p$: E
  (2) An isomorphism $\alpha\colon V_p(E) \tilde{\to} V_0$, given mod K^0
  (3) An isomorphism of the reduction of E with $E_0(\delta)$, $\psi\colon E_{\overline{\mathbf{F}}_p} \tilde{\to} E_0(\delta)$

(4) The diagram

$$\begin{tikzcd} \bigwedge^2 V_p(E) \arrow[r, "\bigwedge^2 \alpha"] \arrow[d, "="] & \bigwedge^2 V_0 \\ d(E)(1) \arrow[r, "\sim", "\psi"'] & d(E_0(\delta))(1) \arrow[u, "\beta(\delta)"'] \end{tikzcd} \tag{1}$$

is commutative.

If K is as above, it amounts to the same to give only mod K, Eq. (1) being required only mod $\det(K)$.

To actually construct $V(K, \delta)$, let us start with

(a) K an open compact subgroup of $\mathrm{GL}(V_0)$
(b) E_0 a supersingular curve up to isogeny over $\overline{\mathbf{F}}_p$ (In practice, $E_0(\delta)$)
(c) $\beta\colon d(E_0) \otimes \overline{\mathbf{Q}}_p(1) \tilde{\to} \bigwedge^2 V_0$ (In practice, $\beta(\delta)$)
(d) $\overline{\beta} = \beta \mod \det(K) \subset \mathbf{Z}_p^\times$ (Only $\overline{\beta}$, not β, will be used)

The construction will involve the auxiliary data of L, a K-stable lattice in V_0.

(a) $\beta^{-1}\left(\bigwedge^2 L\right)(-1)$ is a lattice in $d(E_0) \otimes \mathbf{Q}_p$ and defines an elliptic curve up to p'-isogeny E', iwth $E' \otimes \mathbf{Q} = E_0$. Let me choose an elliptic curve E with level n structure α_n ($n \geq 3$, $(n,p) = 1$) with $E \otimes \mathbf{Z}_{(p)} = E'$. [The construction will be shown to be independent of L, E', E, α_n].

Let M_n be the modular scheme for elliptic curves with level n structure and $e \in M_n(\overline{\mathbf{F}}_p)$ be the point of M_n defined by (E, α_n).

(b) $M(E, \alpha_n)$ *is* the spectrum of the henselization of the local ring at e of $M_n \otimes_Z W(\overline{\mathbf{F}}_p)$.
(c) The completion of $M(E, \alpha)$ is isomorphic to $W(\overline{\mathbf{F}}_p)[\![t]\!]$.
(d) Assume that (E_1, α_n^1) and (E_2, α_n^2) are two systems as above, and that $\phi\colon E_1 \to E_2$ is a p'-isogeny. There is then one and only one isomorphism

$$\overline{\phi}\colon M(E_1, \alpha_n^1) \to M(E_2, \alpha_n^2)$$

fitting in a commutative diagram

$$\begin{tikzcd} \tilde{E}_1 \arrow[dd] \arrow[drr, "\tilde{\phi}"] & & & & E_1 \arrow[llll, hook', dotted] \arrow[dr, "\phi"] \arrow[dd] & \\ & & \tilde{E}_2 \arrow[dd] & & & E_2 \arrow[lll, hook', dotted] \arrow[dd] \\ M(E_1, \alpha_n^1) \arrow[drr, "\overline{\phi}"] \arrow[ddr] & & & & \star \arrow[llll, hook', dotted] \arrow[dr, equal] \arrow[ddl, equal] & \\ & & M(E_2, \alpha_n^2) \arrow[dl] & & & \star \arrow[lll, hook', dotted] \arrow[dll, equal] \\ & W(\overline{\mathbf{F}}_p) & & \star \arrow[ll, hook', dotted] & & \end{tikzcd}$$

In this diagram, $\star$ means $\mathrm{Spec}(\overline{\mathbf{F}}_p)$ (=point), and $\tilde{E}_1$, $\tilde{E}_2$ are the pull-back over $M(E_1, \alpha_n^1)$, $M(E_2, \alpha_n^2)$ of the universal curves over M_n.

This can be better expressed by saying that $M(E_1, \alpha_n^1)$ is the parameter space of the universal deformation of the elliptic curve up to p'-isogeny $E'/\overline{\mathbf{F}}_p$ (universal is with respect to deformation over henselian local $W(\overline{\mathbf{F}}_p)$-algebras). This allows us to write simply $M(E')$ for $M(E_1, \alpha_n^1)$, and $\tilde{E}'$ for the elliptic curve up to p'-isogeny over $M(E')$ defined by (any) E_1.

For n large enough, $K \subset GL(L)$ is the inverse image in $GL(L)$ of a suitable subgroup $\overline{K} \subset \mathrm{GL}(L/p^n L)$.

Let $K(\overline{\mathbf{F}}_p)$ be the field of fractions of $W(\overline{\mathbf{F}}_p)$, and let $\overline{\mathbf{Z}}_p$ be the ring of integers in $\overline{\mathbf{Q}}_p$. The group scheme E'_{p^n} over $M(E')$ is finite étale over $M(E') \otimes K(\overline{\mathbf{F}}_p)$, hence,

$$\underline{\mathrm{Isom}}\left(E'_{p^n}, L/p^n L\right)$$

is a finite étale Galois covering of $M(E') \otimes K(\overline{\mathbf{F}}_p)$, with Galois group $\mathrm{GL}(L/p^n L)$. Let us rather consider

$$\underline{\mathrm{Isom}}_{M(E')\otimes\overline{\mathbf{Q}}_p}\left(E'_{p^n}, L/p^n L\right).$$

This time, we get a *disconnected* covering of $M(E') \otimes \overline{\mathbf{Q}}_p$. A piece of it can be picked as follows: via β, one has $\bigwedge^2 E'_{p^n} = d(E')_p \otimes \mathbf{Z}/p^n(1) = \bigwedge^2 L/p^n L$, and one considers only isomorphisms of "determinant 1". Similarly, $\overline{\beta}$ enables one to pick a component of

$$\overline{K}\backslash\underline{\mathrm{Isom}}_{M(E')\otimes\overline{\mathbf{Q}}_p}\left(E'_{p^n}, L/p^n L\right)$$

We call the component $V_L(K, E_0, \beta)$. It is also the quotient by $\overline{K} \cap \mathrm{SL}(L/p^n L)$ of the chosen component of $\underline{\mathrm{Isom}}_{M(E')}\left(E'_{p^n}, L/p^n L\right) \otimes_{W(\overline{\mathbf{F}}_p)} \overline{\mathbf{Q}}_p$.

Summary. $V_L(K, E_0, \beta)$ depends on $K \subset \mathrm{GL}(V_0)$, a compact open, E_0, up to isogeny over $\overline{\mathbf{F}}_p$, $\beta\colon d(E') \otimes \mathbf{Q}_p(1) \tilde{\to} \bigwedge^2 V_0$, and on a lattice L in V_0, stable by K. It does not depend on the whole of β, but only on $\overline{\beta} = \beta \mod \det(K)$. For $K' \subset K$, one has a map

$$V_L(K', E', \beta) \to V_L(K, E', \beta).$$

The construction of $V_L(K, E_0, \beta)$ is independent of L. More precisely, the system consisting in

(a) $V_L(K, E', \beta)$
(b) the elliptic curve up to isogeny $\tilde{E}_0$ over $V_L(K, E', \beta)$
(c) the universal isomorphism α given mod K: $V_p(\tilde{E}'_0) \to V_0$ (this is deduced from $\overline{\alpha}\colon \tilde{E}'_{p^n} \tilde{\to} L/p^n L$, mod $\overline{K}$, or from $\overline{\alpha}\colon T_p(\tilde{E}') \tilde{\to} L \mod K$.)

is independent of L.

Let L_1 and L_2 be two K-invariant lattices, and let E_1 and E_2 be the corresponding elliptic curves up to p'-isogeny over $\overline{\mathbf{F}}_p$. We are to define an isomorphism

$$\begin{tikzcd} K\backslash\underline{\mathrm{Isom}}\begin{pmatrix} V_p(\tilde{E}_1) & & V_0 \\ \bigcup & , & \bigcup \\ T_p(\tilde{E}_1) & & L_1 \end{pmatrix} \arrow[r, "\sim"] \arrow[d] & K\backslash\underline{\mathrm{Isom}}\begin{pmatrix} V_p(\tilde{E}_2) & & V_0 \\ \bigcup & , & \bigcup \\ T_p(\tilde{E}_2) & & L_2 \end{pmatrix} \arrow[d] \\ M(E_1) & M(E_2) \end{tikzcd}$$

For simplicity, we assume $L_1 \subset L_2$. Then, over the first "Isom", $\tilde{E}_1$ is provided with a subgroup isomorphic to L_2/L_1, call it H. Let $\overline{W}_1(K, E_0, \beta)$ be the normalization of $M(E_1)$ in the first "Isom". Due to the normality of $\overline{W}$ and the fact that elliptic curves have only finitely many subgroup schemes of a given order, the subgroup scheme H of $\tilde{E}_1$ on $\overline{W}_1(K, E_0, \beta) \otimes K(\overline{\mathbf{F}}_p)$ extends as a subgroup scheme H of $\tilde{E}_1$ on $\overline{W}_1(K, E_0, \beta)$. The quotient $\tilde{E}_1/H$ is a deformation of E_2, hence there is a map

$$\overline{W}_1(K, E_0, \beta) \to M(E_2).$$

Further, $V_p(\tilde{E}_1/H) = V_p(\tilde{E}_1)$, and, over $K\backslash\underline{\mathrm{Isom}}\begin{pmatrix} V_p(\tilde{E}_1) & & V_0 \\ \bigcup & , & \bigcup \\ T_p(\tilde{E}_1) & & L_1 \end{pmatrix}$, there is an isomorphism $\alpha \colon V_p(\tilde{E}_1) \to V_0$ carrying $T_p(\tilde{E}_1)$ to L_1 and $T_p(\tilde{E}_1/H)$ to L_2. If W_2 is defined similarly to W_1, one has

$$\begin{tikzcd} K\backslash\underline{\mathrm{Isom}}\begin{pmatrix} V_p(\tilde{E}_1) & & V_0 \\ \bigcup & , & \bigcup \\ T_p(\tilde{E}_1) & & L_1 \end{pmatrix} \arrow[r, dotted] \arrow[d, hook] \arrow[dr] \arrow[ddr] & K\backslash\underline{\mathrm{Isom}}\begin{pmatrix} V_p(\tilde{E}_2) & & V_0 \\ \bigcup & , & \bigcup \\ T_p(\tilde{E}_2) & & L_2 \end{pmatrix} \arrow[d, hook] \\ \overline{W}_1 \arrow[r, dotted] \arrow[d] \arrow[dr, "\text{by } \tilde{E}_1/H"'] & \overline{W}_2 \arrow[d] \\ M(E_1) & M(E_2) \end{tikzcd}$$

This defines the dotted maps which are needed for the isomorphism.

By extending scalars to $\overline{\mathbf{Q}}_p$ and taking one component, one gets the isomorphisms expreseing that $V_L(K, E_0, \beta)$ is independent of L. For $\delta \in D_p$, we note

$$V(K, \delta) = V_L(K, E_0(\delta), \beta(\delta)).$$

Summary. $V(K, \delta)$ is a scheme over $\overline{\mathbf{Q}}_p$; it depends on the compact open K in $\mathrm{GL}(V_0)$ and on $\delta \in D_p$, given modulo multiplication by elements of $\det(K) \subset \mathbf{Z}_p^*$. For K smaller and smaller, the $V(K, \delta)$ form a projective system. Over $V(K, \delta)$ is given an elliptic curve up to isogeny $\tilde{E}$, provided with an isomorphism given mod K $\alpha \colon V_p(\tilde{E}) \tilde{\to} V_0$. In a sense, $\tilde{E}$ is a deformation of $E_0(\delta)$; in particular $d(E_0(\delta)) = d(\tilde{E})$. The isomorphism α is compatible with $\beta(\delta)$:

$$\bigwedge^2 V_p(\tilde{E}) = d(E_0(\delta)) \otimes \mathbf{Q}_p(1) \xrightarrow{\det \alpha = \beta(\delta)} \bigwedge^2 V_0 \qquad \text{mod } \det K$$

In fact, the statement that $\tilde{E}$ is a deformation of E_0 can be made more precise by introducing a suitable $\tilde{E}/\overline{V}(K,\delta)/\overline{Z}_p$.

Variant. Starting with F_0 instead of E_0, one can construct analogues of the $V(K,\delta)$, called $\hat{V}(K,\delta)$, with complete local rings replacing henselian local rings.

3. **Third Construction.** We are interested in the ℓ-adic cohomology groups

$$H^1(V(K,\delta),\mathbf{Q}_\ell)$$

These groups are finite-dimensional, and locally constant as a function of δ (as $V(K,\delta)$ itself is). If K' is a distinguished subgroup of K, one clearly has

$$H^1(V(K,\delta),\mathbf{Q}_\ell) = H^1(V(K',\delta),\mathbf{Q}_\ell)^{K\cap \mathrm{SL}(V_0)/K'\cap \mathrm{SL}(V_0)} \text{ (invariants)}$$

(Note $V(K,\delta)$ depends also only on $K\cap \mathrm{SL}(V_0)$ and δ.)

The *local fundamental object* is the "bundle" $\varkappa/D_p$, with

$$\varkappa_\delta = \varinjlim_K H^1(V(K,\delta),\mathbf{Q}_\ell)$$

There is a notion of "locally constant section of $\varkappa_\delta/D_p$". It is a function $\phi(\delta)$ ($\delta\in D_p, \phi(\delta)\in\varkappa_\delta$) with locally $\phi(\delta)$ in a $H^1(V(K,\delta),\mathbf{Q}_\ell)$ and locally constant. The space of locally constant sections is denoted $\Gamma(D_p,\varkappa)$.

By "transport of structure", the local fundamental object admits actions of:

(a) $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$
(b) $\mathrm{GL}(V_0)$
(c) $\mathrm{Aut}(E_0) = H$

(the actions over D_p being those already described.)

The actions of $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$ and $\mathrm{GL}(V_0)$ are "continuous" (the latter, with respect to the notion of locally constant section).

Proposition. The action of $\mathrm{Aut}(E_0)$ extends as a continuous action of $H(\mathbf{Q}_p)$ on $\varkappa/D_p$.

I don't have a satisfactory proof. My idea of proof would be to go back to the $\overline{W}$ introduced earlier and to express $\varkappa$ in terms of the special fiber of stable models of $\overline{W}$ over suitable ramified extensions of $W(\overline{\mathbf{F}}_p)$.

Intuitively, one may argue that $V(K,\delta)$ and $\hat{V}(K,\delta)$ could have the same cohomology, that $\mathrm{Aut}(F_0) = H(\mathbf{Q}_p)$ acts on $\hat{V}(K,\delta)$, hence on $H^1(\hat{V}(K,\delta),\mathbf{Q}_\ell)$, and that it would be the hell if the action were not continuous [**sic**].

(C). Statement of the local results

(The proofs will be of a global nature.)

It will be easier to work not with $\mathbf{Q}_\ell$-cohomology, but with $\overline{\mathbf{Q}}_\ell$-cohomology, obtained by extending $\mathbf{Q}_\ell$ to an algebraic closure $\overline{\mathbf{Q}}_\ell$. By abuse of language, we will again denote by κ/D_p the "admissible" bundle over D_p, with fiber

$$\kappa_\delta = \varinjlim_K H^1(V(K,\delta),\mathbf{Q}_\ell)\otimes_{\mathbf{Q}_\ell}\overline{\mathbf{Q}}_\ell.$$

The groups $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$, $\mathrm{GL}(V_0)$, $H(\mathbf{Q}_p)$ act admissibly on κ/D_p. The actions on D_p have been computed. Of course, the three actions commute with one another. If $a\in\mathbf{Q}_p^\times$, the actions of the elements $a\in\mathrm{GL}(V_0)$ and $a\in H(\mathbf{Q}_p)$ are *inverse* to one another. In terms of a fixed $\delta_0\in D_p$, this could as well be expressed by saying:

(a) The subgroup of $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p) \times \mathrm{GL}(V_0) \times H(\mathbf{Q}_p)$ formed by the elements such that $d(\sigma)^{-1} \cdot \det(g) \cdot \mathrm{Nrd}(h)^1 = 1$ acts on κ_{δ_0}.
(b) The subgroup $(1, a, a)$ acts trivially.

Mnemotechnik way to check the action on $V(K,\delta)/D_p$, it is good to view a point of $\varprojlim V(K,\delta)/D_p$ as consisting of

(a) $\delta \in D_p$
(b) $\tilde{E}$: elliptic curve up to isogeny on $\overline{\mathbf{Q}}_p$.
(c) $\alpha\colon V_p(\tilde{E}) \tilde{\to} V_0$
(d) ψ, the "specialization map" $\tilde{E} \to \delta(E_0)$

subject to a compatibility between α, ψ and $\beta(\delta)$.

Let $\chi\colon \overline{\mathbf{Q}}_p^\times \to \overline{\mathbf{Q}}_\ell^\times$ be a quasi-character (with open kernel). We denote by $\Gamma(D_p, \varkappa)$ the $\overline{\mathbf{Q}}_\ell$-vector space of locally constant sections of $\varkappa/D_p$, and $\Gamma_\chi(D_p,\varkappa)$ the subspace of those sections for which, for any $a \in \mathbf{Q}_p^\times$ with image $z(a)$ in the center of $\mathrm{GL}(V_0)$, satisfy

$$\begin{aligned} z(a)f &= \chi(a)f, \textit{i.e.}, \\ z(a)f(\delta) &= \chi(a)f(a^2\delta). \end{aligned}$$

Theorem.

(i) $\Gamma_\chi(D_p,\varkappa)$ is a direct sum of triple tensor products

$$\Gamma_\chi(D_p,\varkappa) = \bigoplus_{\phi\in\Phi} V_\phi \otimes V'_\phi \otimes W_\phi,$$

where

(a) V_ϕ is an admissible irreducible representation of $\mathrm{GL}(V_0)$ with central character χ, belonging to the discrete series (= special or supercuspidal)
(b) V'_ϕ is an admissible irreducible (finite-dimensional) representation of $H(\mathbf{Q}_p)$, where the center acts through χ^{-1}
(c) W_ϕ is a 2-dimensional (or 1-dimensional) continuous $\overline{\mathbf{Q}}_\ell$-adic representation of $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$. If it is 2-dimensional, then $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$ acts on $\bigwedge^2 W_\phi$ by the character χ. If W_ϕ is 1-dimensional then the action of $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$ on the space $\bigwedge^2(W_\phi \oplus W_\phi(1))$ again coincides with χ.

(ii) V'_ϕ runs (once and only once) through the representations in (1).
(iii) The same holds for V_ϕ, and V_ϕ and V'_ϕ correspond by the Weil representation construction (suitably normalized)
(iv) W_ϕ is 1-dimensional, if and only if V'_ϕ is also, if and only if V_ϕ is special, and if V'_ϕ is $\mu \circ \mathrm{Nrd}$, then W_ϕ is the character of $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$ defined by $\mu \circ \mathrm{cl}$, where $\mathrm{cl} : W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p) \to \mathbf{Q}_p^\times$ is the map of class field theory.
(v) If V_ϕ is defined by a quasi-character of a quadratic extension of $\overline{\mathbf{Q}}_p$ (with values in $\overline{\mathbf{Q}}_\ell^\times$), then (with a suitable normalization), W_ϕ is induced by that same character.

Except for $p = 2$, this gives a complete description of $\Gamma_\chi(D_p, \chi)$.

For μ a quasi-character of $\overline{\mathbf{Q}}_p^\times$ and $\delta_0 \in D_p$, multiplication by the function $\mu(\delta\delta_0^{-1})$ on D_p provides an isomorphism

$$\Gamma_\chi(D_p, \varkappa) \to \Gamma_{\chi\mu^{-2}}(D_p, \varkappa);$$

if $V_\phi \otimes V'_\phi \otimes W_\phi$ occurs in $\Gamma_\chi(D_p, \varkappa)$, then

$$\left(V_\phi \otimes \mu^{-1} \circ \det\right) \otimes \left(V'_\phi \otimes \mu \circ \mathrm{Nrd}\right) \otimes (W_\phi \otimes \mu \circ \mathrm{cl})$$

occurs in $\Gamma_{\chi\mu^{-2}}(D_p, \varkappa)$.

(D). Global Theory

We consider

K	an open compact subgroup of $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$
$X^\pm$	the Poincaré upper and lower half-plane. That is:
	$X^\pm = \mathrm{Isom}_{\mathbf{R}}\left(\mathbf{Z}^2 \otimes \mathbf{R}, \mathbf{C}\right)/\mathbf{C}^\times \subset \mathrm{Hom}(\mathbf{Z}^2, \mathbf{C})/\mathbf{C}^\times$
	($\mathrm{GL}_2(\mathbf{R})$ acts on the *right* on $\mathbf{Z}^2 \otimes \mathbf{R}$ via its action on $\mathbf{Z}^2 \otimes \mathbf{R} = \mathbf{R}$.)
$M_K^\circ(\mathbf{C})$	$= K\backslash X^\pm \times \mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})/\mathrm{GL}(2, \mathbf{Q})$
k	an integer ≥ 0
μ	the representation Sym^k(dual of obvious representation) of $\mathrm{GL}(2, \mathbf{Q})$
$F_\mu^{\mathbf{Q}}$	the corresponding local system on $M_K^\circ(\mathbf{C})$
$M_K(\mathbf{C})$	the Satake compactification of $M_K^\circ(\mathbf{C})$; $j\colon M_K^\circ(\mathbf{C}) \hookrightarrow M_K(\mathbf{C})$
$F_\mu^{\mathbf{Q}}$	the sheaf $j_* F_\mu^{\mathbf{Q}}$ on $M_K(\mathbf{C})$
$\varkappa$	$= \varinjlim_K H^1(M_K(\mathbf{C}), F_\mu^{\mathbf{Q}})$
	(an admissible representation of $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$ defined over $\mathbf{Q}$)
$\varkappa^{\mathbf{C}}(\mu)$	$= \varkappa(\mu) \otimes \mathbf{C}$
$\varkappa^{\mathbf{Q}_\ell}(\mu)$	$= \varkappa(\mu) \otimes \mathbf{Q}_\ell$
$F_\mu^{\mathbf{Q}_\ell}$	$= F_\mu \otimes \mathbf{Q}_\ell$

One has a decomposition

$$\varkappa^{\mathbf{C}}(\mu) = \varkappa^{k+1,0} \oplus \varkappa^{0,k+1}$$

of $\varkappa^{\mathbf{C}}$ into two complex conjugate subspaces; $\varkappa^{k+1,0}$ is a complex admissible representation of $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$, which can be defined over $\mathbf{Q}$; it is hence isomorphic to the complex conjugate representation $\varkappa^{0,k+1}$. It corresponds to holomorphic modular cusp forms of weight $k+2$ (note that $k + 2 \geq 2$!). For some explicit admissible representation D_{k+2} of $\mathrm{GL}(2, \mathbf{R})$, of the discrete series,

$$\varkappa^{k+1,0} = \mathrm{Hom}_{\mathrm{GL}(2,\mathbf{R})}\left(D_{k+2}, L_0\left(\mathrm{GL}(2, \mathbf{A})/\mathrm{GL}(2, \mathbf{Q})\right)\right).$$

Further, $\{M_K(\mathbf{C})\}$ is naturally defined over $\mathbf{Q}$ (with its $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$-action, under which g maps M_K onto $M_{gKg^{-1}}$), and $F_\mu^{\mathbf{Q}_\ell}$ is an ℓ-adic sheaf, defined over $\mathbf{Q}$. Hence, $\kappa^{\mathbf{Q}_\ell}(\mu)$ carries a $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ action, commuting with $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$, acting by "transport de structure". After extension of $\mathbf{Q}_\ell$ to $\overline{\mathbf{Q}}_\ell$, one has a decomposition

$$\varkappa^{\mathbf{Q}_\ell}(\mu) = \bigoplus_{f \in F}\left(\bigotimes_p V_{f,p}\right) W_f,$$

where the $V_{f,p}$ are irreducible representations of $\mathrm{GL}(2, \mathbf{Q}_p)$, and W_f is a 2-dimensional representation of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$. (Here F is the spectrum of $\mathrm{GL}(2)$).

By Eichler-Shimura-Kuga-Deligne-Ihara-Piatetski-Shapiro-Langlands, if $V_{f,p}$ is of the principal series, then $W_f|_{\mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)}$ (restriction to the decomposition group) is the sum of 2 corresponding characters. If $V_{f,p}$ is special, then $W_f|_{\mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)}$ is the corresponding special ℓ-adic representation of $\mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$.

I can now prove

(A). If $V_{f,p}$ is supercuspidal, then $W_f|_{\mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)}$ is irreducible, and, with the notations of C. Theorem, if $V_{f,p} \sim V_\phi$, then $W_f \sim W_\phi$.

Hence $V_{f,p}$ determines, by a local rule, $W_{f,p} = W_f|_{\mathrm{Gal}(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)}$, and the rule is the obvious one (up to a normalization) if $V_{f,p}$ corresponds by the Weil construction to a character of a quadratic extension.

The method is to use the theory of vanishing cycles to prove that a space to be described below is a quotient of $\varkappa(\mu)\otimes\mathbf{Q}_\ell$ (this is accurate only for $k \neq 0$; I will not bother much about $k = 0$ and the phenomena related to the special representation).

We keep the notation of §B, except that now $V_0 = \mathbf{Q}_p^2$. Let us consider

$$\mathrm{Isom}\left(V_{p'}(E_0), \left(\mathbf{A}^{f,p'}\right)^2\right) \times D_p$$

and the right action of H on it (by composition for the first factor, and the inverse of the already defined action on D_p). On this space, we have the following H-equivariant local system[1]

$$\mathrm{Sym}^k(V_\ell(E)^*) \otimes \varinjlim_K H^1(V(K,\delta), \mathbf{Q}_\ell)$$

(On the second factor, a right action is required; one takes the inverse of the one already constructed.) The space and local system are acted upon by $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$: On the space, $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$ acts by composition on the first factor and acts on D_p in the manner already described. On the local system, $\mathrm{GL}(2, \mathbf{A}^{\mathrm{f}})$ acts trivially on the first factor and acts on the second in the manner already described.

Let H^0 be the space of sections of this local system; this is a representation of $\mathrm{GL}\left(2, \mathbf{A}^{\mathrm{f}}\right)$. There is also an action of $W(\overline{\mathbf{Q}}_p/\mathbf{Q}_p)$ via its action on $V(K,\delta)/D_p$.

I will now relate this H^0 to the spectrum of $H(\mathbf{A})/H(\mathbf{Q})$ and $\Gamma(D_p, \varkappa)$.

For simplicity, let me extend scalars from $\mathbf{Q}_\ell$ to $\mathbf{C}$. (This does not lose the Galois action; to see this, I have to use that the Galois action is continuous for the discrete topology of $\mathbf{Q}_\ell$, which I can do directly.) Let me also choose an isomorphism

$$V_{p'}(E_0) \cong \left(\mathbf{A}^{\mathrm{f},p'}\right)^2,$$

hence

$$H(\mathbf{Q}_\ell) \cong \mathrm{GL}(2, \mathbf{Q}_\ell),\ \ell \neq p.$$

[1]In the original, $V(K,\delta)$ appears without the H^1, but this is inconsistent with his earlier notation; indeed the $V(K,\delta)$ form a projective system as K varies, whereas the $H^1(V(K,\delta), \mathbf{Q}_\ell)$ form an injective system of ℓ-adic local systems on D_p, which is what Deligne wants here.

Then

$$\begin{aligned}
H^0 &= \left\{\begin{matrix}\text{functions } H\left(\mathbf{A}^{\mathrm{f},p'}\right) \to \mathrm{Sym}^k(V_\ell(E)^*) \otimes \Gamma(D_p, \varkappa) \\ \text{with } f(x\gamma) = f(x)\gamma,\ \gamma \in H(\mathbf{Q})\end{matrix}\right\} \\
&= \left\{\begin{matrix}\text{functions } H\left(\mathbf{A}\right) \to \mathrm{Sym}^k(V_\ell(E)^*) \otimes \Gamma(D_p, \varkappa) \\ \text{with } f(g_\infty g_p x\gamma) = g_\infty g_p f(x),\ g_\infty \in H(\mathbf{R}),\ g_p \in H(\mathbf{Q}_p)\end{matrix}\right\} \\
&= \bigoplus_{(\mathbf{A}^\times/\mathbf{Q}^\times)^\wedge} \{\ \}_\chi \\
&= \bigoplus_{\chi} \mathrm{Hom}_{H(\mathbf{R})\times H(\mathbf{Q}_p)}\left(\mathrm{Sym}^k(V)\otimes\Gamma_{\chi_p^{-1}}(D_p,\kappa)^\vee, L_0^\chi(H(\mathbf{A})/H(\mathbf{Q}))\right)
\end{aligned}$$

This can be expressed as follows: In $L_0(H(\mathbf{A})/H(\mathbf{Q}))$, one takes only those elements which at ∞ transform as a vector of a given representation of $H(\mathbf{R})$. The representation of $H(\mathbf{A}^{\mathrm{f}})$ so obtained being written

$$\bigoplus_{f_0\in F_0}\left(\bigotimes_p V_{f_0,p}\right)$$

the following representation occurs in $\kappa(\mu)$:

$$\bigoplus_{f_0\in F}\left(\bigotimes_{\ell\neq p} V_{f_0,\ell}\right)\otimes\bigoplus_{V'_\phi\sim V_{f_0,p}}(V_\phi\otimes W_\phi)$$

Now, by comparison with Jacquet-Langlands §16, plus the fact that supercuspidal representations cannot occur outside $\varkappa(\mu)$ (this is given by Piatetski-Shapiro or Langlands plus the vanishing cycle theory), one gets (a), (b), (ii), (iii), and 2-dimensionality in (c) of Theorem C. One also gets statement (A). By checking this general rule against modular forms attacted to L-functions with grossencharacters of imaginary quadratic fields (and the remark of C), on gets statement (v). The proof of (c) in characteristic 2, in cases not covered by (v), uses entirely different ideas, which I cannot explain here.

Yours sincerely,

P. Deligne.